

KHUVSGUL, Mongolia

WMO: 443860

Lat: 43.6044N

Lon: 109.6361E

Elev: 995

StdP: 89.93

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.1	-24.0	-30.8	0.2	-19.9	-28.4	0.3	-20.2	14.3	-7.6	13.2	-6.3	1.6	320	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	34.9	19.4	33.3	18.8	31.7	18.2	22.8	31.6	21.4	29.9	20.2	28.5	4.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	16.6	28.7	18.5	15.1	26.7	17.2	13.9	25.5	73.5	31.8	67.8	30.1	63.0	28.7	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	12.4	10.6	-28.7	37.9	3.1	1.7	-30.9	39.1	-32.7	40.1	-34.5	41.0	-36.8	42.2		
(5)			-28.9	23.3	3.2	2.9	-31.2	25.3	-33.1	27.0	-34.9	28.6	-37.2	30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-14.3	-10.0	-0.1	8.8	15.8	22.0	25.3	23.3	16.7	7.6	-3.7	-12.6		
	DBStd	14.69	5.17	5.99	6.08	5.48	5.04	3.80	2.98	3.40	4.50	5.17	5.81	5.07		
	HDD10.0	2951	754	561	317	85	9	0	0	0	6	106	412	699		
	HDD18.3	4826	1013	794	572	287	111	11	1	4	81	333	662	958		
	CDD10.0	1728	0	0	3	50	188	361	474	413	206	33	0	0		
	CDD18.3	562	0	0	0	2	31	122	216	158	32	1	0	0		
	CDH23.3	6113	0	0	0	37	437	1325	2419	1593	291	11	0	0		
	CDH26.7	2512	0	0	0	6	149	538	1105	648	65	1	0	0		
(14) Wind	WSAvg	4.3	3.5	3.9	4.7	5.0	5.7	4.5	4.1	3.7	3.8	4.3	4.1	3.9		
(15) Precipitation	PrecAvg	130	2	2	4	3	10	20	32	31	13	7	4	2		
	PrecMax	217	5	12	16	10	43	36	86	78	35	23	12	7		
	PrecMin	79	0	0	0	0	1	2	10	1	1	0	0	0		
	PrecStd	33	1	3	4	3	10	9	15	20	9	6	3	1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.5	8.5	19.2	27.7	33.4	36.2	38.0	36.3	30.6	24.8	13.5	3.6		
		MCWB	-3.2	2.6	7.1	12.4	14.2	19.6	22.3	20.7	16.2	13.3	5.9	-0.3		
	2%	DB	-1.5	5.1	15.5	24.1	30.5	33.8	35.4	34.1	28.6	21.3	10.2	-0.6		
		MCWB	-4.9	0.3	5.4	10.5	13.9	18.0	20.2	19.8	16.1	11.3	4.2	-3.7		
	5%	DB	-4.3	2.2	12.6	21.1	28.0	31.9	33.7	32.1	26.8	18.6	7.1	-3.1		
		MCWB	-6.9	-2.0	3.8	9.2	13.0	17.5	19.6	18.9	15.0	9.7	2.1	-5.7		
	10%	DB	-6.6	-0.6	9.8	18.5	25.5	29.8	32.0	30.3	24.8	15.9	4.6	-5.2		
		MCWB	-8.7	-4.2	2.2	7.8	12.0	16.4	19.2	18.0	13.9	8.0	0.4	-7.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.8	3.6	8.3	14.4	17.6	22.8	26.5	23.6	20.2	15.0	6.6	-0.6		
		MCDB	1.3	7.8	16.0	25.6	28.9	34.6	36.4	33.1	28.0	22.7	12.5	3.5		
	2%	WB	-4.9	0.3	5.8	11.5	15.3	20.5	23.7	22.1	18.1	12.4	4.5	-3.5		
		MCDB	-1.6	4.8	15.1	21.4	26.2	30.1	32.3	30.3	25.9	19.4	9.9	-0.7		
	5%	WB	-6.8	-1.8	4.1	9.9	14.0	19.2	22.1	20.9	16.5	10.1	2.5	-5.6		
		MCDB	-4.3	2.3	11.9	19.5	25.7	28.2	30.4	29.7	23.1	17.5	7.0	-3.3		
	10%	WB	-8.6	-3.9	2.4	8.4	12.8	18.1	20.9	19.6	15.1	8.5	0.7	-7.2		
		MCDB	-6.6	-0.4	9.4	18.1	23.7	26.8	28.9	27.8	22.7	15.1	4.6	-5.2		
(35) Mean Daily Temperature Range	MDBR		10.6	12.1	13.0	12.5	13.0	11.5	10.9	11.1	11.6	11.5	10.4	9.6		
	5% DB	MCDBR	12.3	14.7	15.6	15.8	15.7	13.7	12.9	12.9	13.5	13.6	12.6	11.3		
		MCWBR	9.8	10.8	9.5	8.8	7.7	6.2	5.1	5.8	6.6	7.9	8.4	8.8		
	5% WB	MCDBR	12.2	14.4	15.2	14.5	14.7	11.7	11.1	11.6	11.7	13.1	12.3	11.0		
		MCWBR	9.8	10.8	9.6	8.9	8.5	6.4	6.2	6.7	7.3	8.5	8.6	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.274	0.304	0.346	0.398	0.438	0.462	0.437	0.416	0.382	0.334	0.292	0.273		
	taud		2.265	2.190	2.104	2.011	1.953	1.976	2.113	2.148	2.168	2.236	2.263	2.244		
	Ebn at Noon		852	881	883	862	836	815	832	837	835	833	817	811		
	Edh at Noon		85	110	138	165	181	177	153	143	129	104	85	79		
(44) All-Sky Solar Radiation	RadAvg		2.22	3.33	4.91	6.17	6.94	6.86	6.60	6.07	5.08	3.82	2.55	1.88		
	RadStd		0.26	0.31	0.27	0.28	0.31	0.33	0.28	0.24	0.20	0.15	0.16	0.20		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days					
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)		
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
(47)	Regional (5 neighbors)	N/A	N/A	-1.14	N/A	-0.57	-0.75	N/A	N/A	N/A	N/A	(47)		

Nomenclature: See separate page